

Phase 3: the bottom bar

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Five buttons on the bottom bar: Reposition, Spare tire, Repair, Fuel and Lights. Right now they say what they will cost and then do nothing. Phase 3 makes them work.

This is a small phase next to phase 2. Most of the hard parts are already built.

1 What already exists

Shipped in phase 1 and 2, so it is not work in this phase:

- The five buttons, with keys bound to each one.
- The rule that the car must be stopped.
- The penalty system. Adding seconds to a run already works.
- The results screen already names each penalty by its reason: Recovery, Spare tire, Repair.
- Penalties only exist inside a run. Free driving costs nothing.

What is missing is the action itself and the wait that goes with it.

2 The five buttons

Button	Hold	Penalty	What it does
Reposition	5s	60s	Puts the car back upright and on the road.
Spare tire	30s	30s	Fixes flat tires only.
Repair	60s	30s	Fixes the whole car.
Fuel +25%	20s	none	Adds a quarter of a tank.
Lights	none	none	Head lights. Works already.

The holds and penalties come from the server config, so you can change any number without anyone reinstalling the mod.

Hold means the car is locked in place for that long while the job is done. **Penalty** means seconds added to your race time at the end.

The race clock keeps running during a hold. So a spare tire in a race costs you 30 seconds of waiting and 30 seconds on the clock. One minute in total.

3 What happens when you press one

1. The client checks you are in a car and stopped. If not, it says so and nothing happens.
2. The client asks the server.
3. The server decides. It checks you are in a run, that you are not already holding, and that the button is one it knows.
4. If yes, the server starts the hold and adds the penalty straight away.

5. The client freezes the car and shows a countdown.
6. When the hold ends, the client does the actual job in the game and unfreezes the car.

The penalty is added at the start, not the end. If it were added at the end, you could quit the hold and get the repair for free.

4 Who decides what

The server owns the decision, the hold clock and the penalty. It is the same rule as the rest of the project: anything a player could gain by lying lives on the server.

The client only does what the game needs done, after the server says yes. It never tells the server that a job finished, because the server already knows when the hold ends.

5 Rules and edge cases

- You cannot press a second button while a hold is running.
- You cannot cancel a hold. The penalty is already counted.
- Leaving the server during a hold ends the run the normal way, as a did not finish.
- If the game cannot do the job, for example a car with no fuel tank, the player is told plainly and the penalty is given back.
- Outside a run there is no penalty and no hold. The buttons just work.
- Lights need no hold and no stop, so they stay instant.
- Nothing here adds work to the per frame loop. The hold is one number compared on the tick that already runs.

6 On screen

- A countdown while the hold runs, in the same place as the race clock.
- The clock keeps counting during the hold, because it is still costing you time.
- The penalty shows in the count next to the clock the moment it lands, the same as a missed checkpoint does now.
- Buttons grey out while a hold is running.

7 Questions, with a default if you do not mind

Each of these has a default. Skip any you are happy with.

1. **Does the hold count on top of the penalty?** Default: yes. The hold is the wait, the penalty is the punishment. The other option is that the hold replaces the penalty.
2. **Where does Reposition put the car?** Default: back at the last checkpoint you passed, upright and facing the right way. The other option is upright where it stands.
3. **Should there be a limit per run?** Default: no limit. The penalties are the limit.
4. **Should Repair be allowed during a race at all?** Default: yes, at 60 seconds of waiting and 30 on the clock.

5. **Holds outside a race?** Default: no. Free driving costs nothing and waits for nothing.

There is also a speed limit setting sitting in the config, 37 mph, that nothing uses yet. If that was meant to be a pit lane or service area limit, tell me and I will say whether it belongs in this phase or a later one.

8 Order of work

1. Server side first: the hold state, the checks, the penalty. All of it testable without the game.
2. The four actions in the game, one at a time, easiest first: fuel, tires, repair, reposition.
3. The countdown and the greyed out buttons.
4. Testing in game, one button at a time, in a race and out of one.

Server first means most of the phase is proven before the game is opened. That is what worked in phase 2.

9 Testing

- Mock tests for the hold: start, block a second press, end on time, penalty added once and not twice.
- A test that a hold survives the run ending badly, so nothing is left stuck.
- In game, each button twice: once in a race and once in free driving.
- One run using every button, then read the results screen and check every penalty is listed with the right reason.

10 What could go wrong

- **The game may not repair tires on their own.** If there is no clean way to fix only the tires, Spare tire becomes a full repair with the lower cost, and I will say so rather than pretend.
- **Fuel differs by vehicle.** Electric cars have no tank. Those get told plainly instead of failing in silence.
- **Freezing a car in multiplayer.** Other players will see the car stopped, which is correct, but it needs checking with two drivers.

11 Not in this phase

Qualifying and Race on a shared grid are phase 4. Records and personal bests are phase 5. CoPilot and Challenges are phase 6.

Phase 2 is delivered and running on the server. Phase 3 starts on your word, and the five questions above are the only thing that would change the shape of it.